

Bacterial Peptidoglycan Polymerization and Crosslinking in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00550 "Peptidoglycan biosynthesis" — resolved to the reusable module *Bacterial peptidoglycan polymerization and crosslinking* (lipid II-downstream) **Module area:** lipid_cell_envelope_metabolism

1. Executive Summary

The lipid II-downstream peptidoglycan (PG) polymerization/crosslinking module is **fully satisfiable** in *P. putida* KT2440. Every canonical step class of the generic module maps to at least one candidate gene whose role is corroborated by conserved Pfam domain architecture: septal SEDS-bPBP synthesis (FtsW/FtsI), lateral-wall SEDS-bPBP synthesis (MrdB/RodA plus PBP2/MrdA), class-A bifunctional PBP synthesis (PBP1A/MrcA, PBP1B/MrcB, PbpC), monofunctional glycan polymerization (MtgA), and pentapeptide stem trimming (DacA/PBP5 D,D-carboxypeptidase). No canonical step is a genuine gap in the target organism.

However, the KEGG-derived candidate list requires substantial curation before it can drive automated module satisfiability. Three structural issues dominate: (i) **scope contamination** — only 11 of the 23 candidate genes actually belong to the polymerization/crosslinking module; the remaining 12 are cytoplasmic Mur/Ddl precursor-synthesis enzymes or undecaprenyl-carrier metabolism enzymes that the module scope explicitly excludes; (ii) **paralog multiplicity** — KT2440 carries **two genuine PBP2/MrdA D,D-transpeptidase paralogs** (PP_3741 and PP_4807, 73.5% identical), which must both be represented rather than collapsed; and (iii) **annotation completeness/misassignment** — pbpC has an incomplete EC (only its glycosyltransferase activity is captured, not its transpeptidase), and dacB (PP_2098) is a PBP4/peptidase-S13 D,D-endopeptidase that does *not* satisfy the DacA/PBP5 carboxypeptidase step and should not be conflated with it.

Two module-revision needs emerged. First, the generic module has no node for the **PBP4-class D,D-endopeptidase** activity that dacB encodes, which is functionally relevant in *Pseudomonas* β -lactam/AmpC biology. Second, the module omits **L,D-transpeptidation** (3-3 crosslink formation) entirely; a direct proteome scan confirms KT2440 encodes two YkuD-family L,D-transpeptidases (PP_1451 and PP_2320) that are absent from the candidate list — a documented module gap rather than an organism gap. Critically, **all in-scope genes are homology-inferred (UniProt evidence level PE=3) with no KT2440-specific experimental characterization**, so all functional assignments carry species-transfer uncertainty. The two MrdA/PBP2 paralogs should be flagged `candidate_uncertain` and promoted to full gene review.

2. Target-Organism Pathway Definition

P. putida KT2440 is a Gram-negative γ -proteobacterium with DAP-type (A1 γ) peptidoglycan, so its PG polymerization/crosslinking machinery follows the canonical two-machine (divisome + elongasome) plus class-A PBP architecture.

What is included

The module covers the **membrane-associated, lipid II-downstream reactions** that build and crosslink the sacculus:

- **Glycan-chain polymerization (glycosyltransfer, EC 2.4.99.28):** transfer of the disaccharide-pentapeptide from lipid II onto growing glycan strands, performed by two structurally unrelated enzyme classes — **SEDS-family polymerases** (FtsW, RodA/MrdB; Pfam PF01098) and the **GT51/Transglycosylase domains of class-A PBPs** plus the standalone monofunctional transglycosylase MtgA (Pfam PF00912). SEDS enzymes add lipid II monomers to the reducing end of the strand ([PMID: 31386359](#)).
- **Peptide crosslinking (D,D-transpeptidation, 4→3 crosslink; EC 3.4.16.4):** by **class-B PBPs (bPBPs)** — FtsI/PBP3 (division) and MrdA/PBP2 (elongation) — and by the transpeptidase (PF00905) domains of class-A PBPs.
- **Stem-peptide maturation: D,D-carboxypeptidase** trimming of the terminal D-Ala from pentapeptide stems (serine-type, S11/PBP5; DacA) to control crosslink density.

What must be kept separate (neighboring pathways)

The module scope statement explicitly excludes, and this review confirms should be curated out of this bucket:

- **Cytoplasmic precursor synthesis** — the Mur ligase pathway (MurA, MurB, MurC, MurD, MurE, MurF), MraY, MurG, and the D-Ala–D-Ala ligases (DdlA, DdlB), which build UDP-MurNAc-pentapeptide and lipid II *before* export (KEGG ppu00300/00470/01502).
- **Undecaprenyl-carrier metabolism** — UppS (undecaprenyl-diphosphate synthase; ppu00900) and UppP (undecaprenyl-diphosphatase; ppu00552), which regenerate the lipid carrier.
- **Lipid II flipping** (MurJ/Amj) and **PG recycling** (AmpD/AmpG/NagZ/Mpl) — both explicitly outside scope.

Alternate names / database definitions

- KEGG map **ppu00550** ("Peptidoglycan biosynthesis") is a *whole-map* bucket, broader than this mechanistic module — the reason 12 of the 23 candidates are out-of-scope.
- The machinery is referred to as the **divisome** (FtsW-FtsI), the **elongasome / Rod system** (RodA-PBP2), the **class-A PBPs** (aPBPs), and collectively as **penicillin-binding proteins (PBPs)** / "murein synthases."
- GO anchors: peptidoglycan glycosyltransferase activity (GO:0008955); serine-type D-Ala-D-Ala carboxypeptidase activity (GO:0009002); peptidoglycan biosynthetic process (GO:0009252).

3. Expected Step Model and Satisfiability

The generic module defines five step classes. All are covered in KT2440:

Module step	Enzyme class	KT2440 candidate gene(s)	Status
1. Septal glycan polymerization	FtsW SEDS GTase	ftsW PP_1336	Covered
1. Septal D,D-transpeptidation	FtsI/PBP3 bPBP	ftsI PP_1331	Covered
2. Lateral glycan polymerization	RodA/MrdB SEDS GTase	mrdB PP_4806	Covered
2. Lateral D,D-transpeptidation	MrdA/PBP2 bPBP	mrdA-II PP_4807 (operonic) + mrdA-I PP_3741 (paralog)	Covered (2 paralogs)
3. Bifunctional GTase + TPase	Class-A PBPs	mrcA PP_5084, mrcB PP_4683, pbpC PP_0572	Covered
4. Monofunctional glycan polymerization	Standalone GTase	mtgA PP_5107	Covered
5. Pentapeptide stem trimming	DacA/PBP5 D,D-CPase	dacA PP_4803	Covered

Steps missing from the generic module but present in the organism (module gaps, not organism gaps):

- **PBP4-class D,D-endopeptidase** — **dacB** (PP_2098) is a peptidase-S13/PBP4 enzyme with D,D-endo/carboxypeptidase activity; the generic module has no node for it.
- **L,D-transpeptidation (3-3 crosslink)** — encoded by YkuD-family PP_1451 and PP_2320, absent from the candidate list and unmodeled.

Steps probably not expected / not applicable: None of the five canonical steps is absent. There is no evidence for loss of any elongasome or divisome component in KT2440.

4. Candidate Genes and Evidence

All 11 in-scope assignments below are supported by **conserved Pfam domain architecture** (verified from UniProt) and pathway-database placement; **none has direct KT2440 experimental characterization** (all UniProt PE=3, inferred from homology). Paralog relationships were computed by global alignment.

4.1 In-scope genes (11) — the actual module

Gene	Locus	UniProt	Length	Pfam architecture	Module role	Curation note
ftsW	PP_1336	Q88N77	—	PF01098 (10 TM)	Septal SEDS GTase	High confidence; divisome partner of FtsI
ftsI	PP_1331	Q88N82	—	PF03717 + PF00905, 1 TM	Septal bPBP (PBP3) TPase	~38–39% id to PBP2; aztreonam/ceftazidime target
mrdB/rodA	PP_4806	Q88DL9	—	PF01098 (9 TM)	Lateral SEDS GTase	Operonic upstream of mrdA-II; Rod system
mrdA-II	PP_4807	Q88DL8	629 aa	PF03717 + PF00905, 1 TM	Lateral bPBP (PBP2) TPase	Canonical operonic elongasome PBP2
mrdA-I	PP_3741	Q88GI2	631 aa	PF03717 + PF00905, 1 TM	Lateral bPBP (PBP2) TPase	Second, dispersed PBP2 paralog
mrcA	PP_5084	Q88CU6	—	PCB_OB + Transgly + Transpeptidase	Class-A PBP1A	Bifunctional; EC 2.4.99.28 + 3.4.16.4 (correct)
mrcB	PP_4683	Q88DY5	—	Transgly + Transpeptidase + UB2H	Class-A PBP1B	No EC listed — add bifunctional EC pair
pbpC	PP_0572	Q88QC2	784 aa		Class-A PBP1C	

Gene	Locus	UniProt	Length	Pfam architecture	Module role	Curation note
				PF00912 + PF00905 + PF06832		EC 3.4.16.4 TPase not captured
mtgA	PP_5107	Q88CS3	236 aa	PF00912 only	Monofunctional GTase	Accessory/ dispensable
dacA	PP_4803	Q88DM2	—	PF00768 (S11) + PF07943	DacA D,D- carboxypeptidase	Satisfies stem- trimming step

SEDS glycosyltransferases (ftsW, mrdB). Both carry PF01098 with the expected transmembrane counts (10 TM and 9 TM). FtsW is the septal SEDS polymerase partnering FtsI; MrdB/RodA sits immediately upstream of mrdA-II in the canonical *mrd/rod* operon and is the lateral-wall SEDS polymerase of the Rod system/elongasome. Transfer from *E. coli* and general bacterial SEDS biology is **strong** because these are universally conserved, essential enzymes.

PBP2 paralogs (Finding F002). A global Needleman–Wunsch alignment gives **73.5% identity** between MrdA-I (PP_3741, 631 aa) and MrdA-II (PP_4807, 629 aa). Both share the canonical PBP2 architecture (PF03717 PBP_dimer + PF00905 Transpeptidase, single N-terminal TM anchor) and are only ~38–39% identical to FtsI/PBP3 (PP_1331), confirming both are **elongasome-type PBP2, not divisome FtsI**. PP_4807 sits immediately downstream of mrdB/rodA (the "housekeeping" elongation PBP2); PP_3741 is a genome-dispersed second copy. This is a genuine gene duplication, not a mis-split annotation. The physiological division of labor (redundant, specialized, or condition-specific) is **unknown in KT2440** — these are the strongest candidates for `candidate_uncertain`.

Class-A bifunctional PBPs (mrcA, mrcB, pbpC). MrcA/PBP1A is annotated with both EC 2.4.99.28 and 3.4.16.4 — a correctly-annotated bifunctional template. MrcB/PBP1B (Transgly + Transpeptidase + UB2H/PF14814) is unambiguously bifunctional by architecture but carries **no EC in the metadata**. PbpC/PBP1C (PF00912 + PF00905 + PF06832 BiPBP_C) is a full class-A PBP whose EC lists **only the glycosyltransferase (2.4.99.28)**, omitting the transpeptidase (Finding F003). In *E. coli*, PBP1C is a minor/low-activity synthase, so exact functional transfer to KT2440 is **uncertain** even though the structural call is clear.

Monofunctional transglycosylase (mtgA). PF00912-only, 236 aa, EC 2.4.99.28 — an accessory glycan polymerase, individually dispensable in model bacteria. Activity transfer is **strong**; physiological necessity is **moderate**.

D,D-carboxypeptidase (dacA). PF00768 (Peptidase_S11) + PF07943 (PBP5_C), EC 3.4.16.4 — the LMM D,D-carboxypeptidase (PBP5) that satisfies the module's stem-trimming step. Transfer **strong**.

4.2 Out-of-scope genes (12) — should be removed from this module

Correctly PG-related but belonging to **cytoplasmic precursor synthesis** or **carrier metabolism**, both explicitly excluded by the module scope (Finding F001):

- **Mur ligases / precursor:** murA (PP_0964), murB (PP_1904), murC (PP_1338), murD (PP_1335), murE (PP_1332), murF (PP_1333), murG (PP_1337)
- **First membrane step (pre-export):** mraY (PP_1334)
- **D-Ala–D-Ala ligases:** ddlA (PP_4346), ddlB (PP_1339)
- **Undecaprenyl carrier:** uppS (PP_1595), uppP (PP_2862)

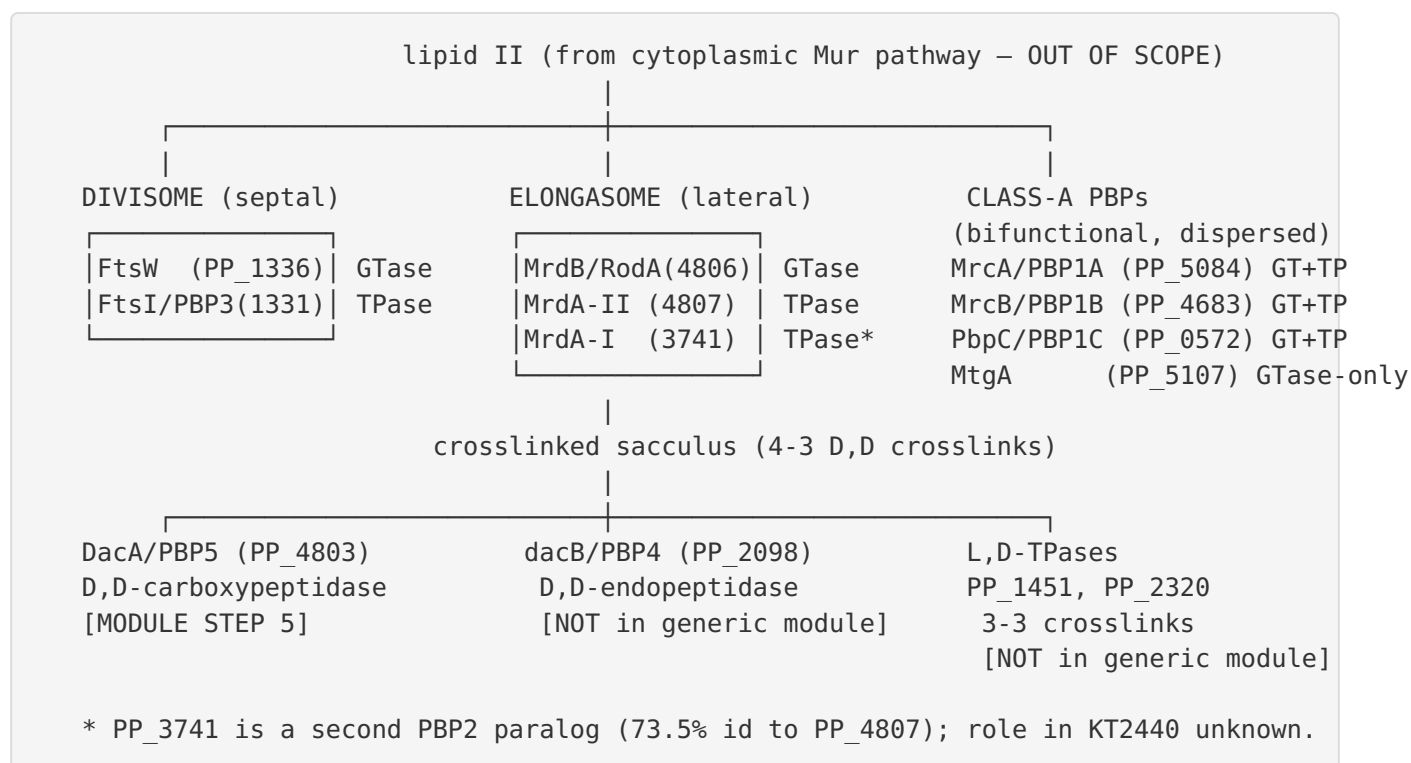
These should be reassigned to their correct buckets rather than counted toward module satisfiability.

4.3 dacB — distinct enzyme, not the DacA step (Findings F003, F004)

dacB (PP_2098, Q88L37, 470 aa) carries Pfam **PF02113 (Peptidase_S13)**, the **PBP4 family** (D,D-endopeptidase / D,D-carboxypeptidase). This is biochemically and structurally distinct from **dacA** (PP_4803, PF00768/Peptidase_S11, PBP5-type). The generic module's single "DacA D,D-carboxypeptidation" step is satisfied by **dacA**, not **dacB**. **dacB** represents an additional, unmodeled activity that is functionally important in *Pseudomonas* — LMW-PBP (PBP4/dacB) inactivation drives AmpC β -lactamase overexpression in *P. aeruginosa* (PMID: 41206063). A pipeline that maps **dacB** to the DacA step by the generic "D,D-carboxypeptidase" label would be in error.

5. Mechanistic Model / Interpretation

The KT2440 PG polymerization/crosslinking machinery follows the canonical Gram-negative two-machine architecture, with lineage-specific paralog expansion at PBP2:



The interpretation for curation is that KT2440's genome is **complete for canonical PG polymerization/crosslinking**, and in fact **richer** than the generic module in two respects (a PBP2 duplication, and both PBP4/S13 endopeptidase and YkuD L,D-transpeptidase activities). The module boundaries are correct in excluding precursor synthesis and carrier metabolism, but the module under-represents the LMW-PBP / non-classical crosslinking layer that is biologically and clinically important in *Pseudomonas*.

6. Gaps, Ambiguities, and Likely Over-Annotations

6.1 Scope contamination (highest-impact curation issue)

Only **11 of 23** KEGG ppu00550 candidates are in-scope. The KEGG map bundles the whole PG pathway, so the 12 Mur/Ddl/MraY/Upp genes are propagated into this bucket incorrectly, inflating apparent satisfiability. Correcting this is the single most impactful curation action.

6.2 Paralog ambiguity (not over-propagation)

The two PBP2/MrdA paralogs (PP_3741, PP_4807) are **real** (73.5% identity, correct Pfam, one operonic). The generic module already anticipates "MrdA-I/MrdA-II" — this is confirmed, not spurious. Neither should be dropped; PP_3741's specific role is the key open question.

6.3 Broad / incomplete EC and GO mappings

- **pbpC (PP_0572)**: under-annotated — EC 3.4.16.4 transpeptidase missing despite PF00905.
- **mrcB (PP_4683)**: no EC despite unambiguous class-A bifunctional architecture.
- **dacB (PP_2098)**: generic "D-alanyl-D-alanine carboxypeptidase" name with no EC; its S13/PBP4 endopeptidase identity is not captured — risk of conflation with the DacA step.
- **mrcA (PP_5084)**: dual EC (2.4.99.28 + 3.4.16.4) correctly applied — the template for bifunctional PBP annotation.

6.4 Missing module nodes (module_needs_revision)

- **L,D-transpeptidation (3-3 crosslink)**: direct proteome evidence (Finding F005) — a scan of UP000000556 for Pfam **PF03734 (YkuD/L,D-transpeptidase)** returns exactly two proteins, **PP_1451 (Q88MW7)** and **PP_2320 (Q88KH0)**, both "L,D-TPase catalytic domain-containing protein." Neither is in the candidate list. L,D-transpeptidases produce 3-3 crosslinks and contribute to β -lactam tolerance and cell-wall remodeling in Gram-negative bacteria; the module has no node for them. This is a **module gap, not an organism gap**.
- **PBP4 D,D-endopeptidase (dacB)**: no node exists for this activity.
- **Additional LMM-PBPs**: *Pseudomonas* genomes typically encode further PBP5-like/PBP7-like carboxy/endopeptidases; the single DacA slot under-represents this layer.

6.5 Naming inconsistency (Finding F005)

A name search for "penicillin-binding" in the KT2440 proteome returns only 5 proteins (mrcA, mrcB, mrdA-I, mrdA-II, ftsI). The SEDS polymerases (ftsW, mrdB), the class-A PbpC, MtgA, and the LMW PBPs (dacA, dacB) are **not retrievable by the "penicillin-binding" name string** — curators relying on name-based retrieval will miss functionally central genes. Retrieval must be by Pfam/EC, not name.

7. Evidence Base

All functional assignments for KT2440 in-scope genes rest on **homology and domain architecture (UniProt PE=3)**; there is no direct experimental characterization of these enzymes in KT2440 itself. Supporting mechanistic literature (from related organisms) and its transfer strength:

PMID	Finding supported	Organism / transfer strength
31386359	RodA/FtsW are SEDS PG polymerases (verified quote: "the ubiquitous Shape, Elongation, Division, and Sporulation (SEDS)-family proteins RodA and FtsW were shown to be peptidoglycan polymerases")	General bacterial; strong — SEDS mechanism conserved; KT2440 ftsW/mrdB carry PF01098
33558391	Lateral RodA-PBP2 elongasome (verified quote: "PG is incorporated along the cell cylinder by the RodA-PBP2 synthase of the multi-protein Rod system (elongasome)")	General bacterial; strong — KT2440 has the mrdB-mrdA operon
28289035	PBP2/PBP3 as essential β -lactam/enhancer targets in <i>Pseudomonas</i>	<i>P. aeruginosa</i> ; moderate transfer to <i>P. putida</i> (same genus, conserved PBPs)
41206063	PBP4/dacB is a distinct LMW-PBP whose inactivation drives AmpC induction	<i>P. aeruginosa</i> PAO1; moderate — supports treating dacB separately from dacA
33830599	<i>P. putida</i> cell-wall chemical editing; PG crosslinking/transpeptidase biology	<i>P. putida</i> (direct genus/species context) + Rhizobiales; moderate–strong context
28861525	PBP2 inactivation alters muropeptide profile and β -lactamase expression	<i>Pseudomonas</i> ; moderate — supports functional importance of PBP2
42294650	PBP2 mutation (V516M) confers zidebactam resistance — PBP2 is the DBO target	<i>Pseudomonas</i> ; moderate — supports PBP2 as elongation TPase

Papers on xeruborbactam (PMID: 36102663), avibactam/AmpC (PMID: 33802668), and KT2440 Tn-seq/BarSeq metabolism screens (PMID: 40302248, PMID: 32826213) provide broader context but do not directly characterize the KT2440 PG synthases.

8. Module and GO-Curation Recommendations

Per-step module status:

Module step	Recommended status	Gene(s)
Septal SEDS GTase (FtsW)	covered	ftsW PP_1336
Septal bPBP TPase (FtsI)	covered	ftsI PP_1331
Lateral SEDS GTase (RodA/MrdB)	covered	mrdB PP_4806
Lateral bPBP TPase (MrdA/PBP2)	covered ; flag paralogs candidate_uncertain	mrdA-II PP_4807, mrdA-I PP_3741
Class-A PBP1A	covered	mrcA PP_5084
Class-A PBP1B	covered (add missing EC)	mrcB PP_4683
Class-A PbpC	covered structurally (add EC 3.4.16.4)	pbpC PP_0572
Monofunctional GTase (MtgA)	covered (accessory)	mtgA PP_5107
DacA D,D-carboxypeptidase	covered	dacA PP_4803
PBP4 D,D-endopeptidase	module_needs_revision (new node)	dacB PP_2098
L,D-transpeptidation (3-3)	module_needs_revision / new companion module	PP_1451, PP_2320
12 Mur/Ddl/MraY/Upp genes	remove from bucket (out of scope)	—

Actions: 1. **Remove** the 12 out-of-scope precursor/carrier genes from the polymerization module bucket; re-map to precursor synthesis / isoprenoid-carrier modules. 2. **Complete EC annotations** for pbpC (add EC 3.4.16.4) and mrcB (add EC 2.4.99.28 + 3.4.16.4), using mrcA as the correctly-annotated template. 3. **Reclassify dacB** as PBP4/peptidase-S13 D,D-endopeptidase; do not use it to satisfy the DacA carboxypeptidase step. Add a module node for D,D-endopeptidase activity. 4. **Add a module node / GO representation for L,D-transpeptidation** (3-3 crosslink) and associate PP_1451/PP_2320. A GO term request may be warranted if the module ontology lacks a 3-3 crosslink node. 5. **Keep SEDS GTase (GO:0008955 via PF01098) distinct from GT51 class-A GTase** in curation, and ensure GO:0009002 is applied only to the S11/PBP5 DacA family. 6. **Flag the naming inconsistency:** retrieval of PG synthases must be by Pfam/EC, not by the "penicillin-binding" name string.

9. Genes to Promote to Full fetch-gene Review

Priority for full gene-level review (all are PE=3, no direct KT2440 data):

1. **mrda-I (PP_3741)** and **mrda-II (PP_4807)** — the two PBP2 paralogs. Highest priority: paralog role assignment, operon context, redundancy vs specialization. Flag candidate_uncertain.
2. **pbpC (PP_0572)** — resolve bifunctional EC/GO annotation; assess whether it is an active TPase in KT2440.
3. **dacB (PP_2098)** — confirm PBP4/S13 identity and D,D-endopeptidase activity; decide module placement.
4. **mrcB (PP_4683)** — add missing bifunctional EC.
5. **PP_1451 and PP_2320** — YkuD L,D-transpeptidases; promote to establish the L,D-crosslink node even though absent from the original candidate list.
6. **ftsW (PP_1336)** and **mrdb (PP_4806)** — confirm SEDS partner pairing (FtsW–FtsI vs RodA–PBP2) as reference anchors.

10. Limitations and Knowledge Gaps

- **No direct experimental evidence in KT2440.** Every in-scope functional assignment is homology-inferred (UniProt PE=3). Mechanistic confidence rests on transfer from *E. coli*/

general bacterial SEDS-bPBP biology and from *P. aeruginosa* PBP studies. Transfer within genus (*P. aeruginosa* → *P. putida*) is moderate; transfer from distant taxa is weaker.

- **Paralog function is unresolved.** The distinct roles of the two PBP2 paralogs (PP_3741 vs PP_4807) cannot be assigned from sequence alone.
- **L,D-transpeptidase and PBP4 activities are inferred from domain content only;** their in vivo contribution to KT2440 crosslinking has not been measured.
- **Module ontology limits.** Whether suitable GO terms / module nodes exist for 3-3 L,D-crosslinking and PBP4 endopeptidase activity was not verified against the live ontology.
- This review analyzed sequence/domain metadata and literature; **no muropeptide (HPLC/MS) crosslink profiling** of KT2440 was available to confirm which crosslink types are actually made.

11. Proposed Follow-up Experiments / Actions

1. **Muropeptide profiling** of KT2440 sacculi (LC-MS) to quantify 4-3 vs 3-3 crosslinks and confirm L,D-transpeptidase activity in vivo.
2. **Single and double knockouts** of PP_3741 / PP_4807 to resolve PBP2 paralog redundancy and essentiality (with Bocillin-FL PBP profiling).
3. **Bocillin-FL PBP labeling** of KT2440 membranes to enumerate expressed PBPs and cross-check the in-silico roster.
4. **Curation task tickets:** (a) strip 12 precursor/carrier genes from bucket; (b) complete pbpC/mrcB EC; (c) reclassify dacB; (d) add L,D-TPase + PBP4 nodes; (e) add PP_1451/PP_2320.
5. **Expert question:** Does the curation ontology already have GO/module nodes for L,D-transpeptidation (3-3 crosslink) and PBP4 D,D-endopeptidase? If not, file GO term requests.

12. Key References

- Welsh MK, et al. *Direction of Chain Growth and Substrate Preferences of SEDS-Family Peptidoglycan Glycosyltransferases*. PMID: 31386359 — RodA/FtsW are the SEDS PG polymerases.

- Rohs PDA, *et al.* Identification of potential regulatory domains within the MreC and MreD components of the cell elongation machinery. PMID: 33558391 — RodA–PBP2 synthase of the Rod system (elongasome).
- Moya B, *et al.* WCK 5107 (Zidebactam)/WCK 5153: novel PBP2 inhibitors with β -lactam enhancer activity against *P. aeruginosa*. PMID: 28289035 — PBP2/PBP3 as essential *Pseudomonas* targets.
- Unravelling the triad of PBPs, β -lactamase activity, and mRNA dynamics in *P. aeruginosa* AmpC induction. PMID: 41206063 — PBP4/dacB as a distinct LMW-PBP regulator/enzyme.
- Aliashkevich A, *et al.* d-canavanine affects peptidoglycan structure, morphogenesis and fitness in *Rhizobiales*. PMID: 33830599 — uses *P. putida*; PG crosslinkage and transpeptidase editing.
- Characterizing the V516M penicillin-binding protein 2 (PBP2) mutation causing zidebactam resistance. PMID: 42294650 — PBP2 as elongation transpeptidase target.
- Impacts of Penicillin Binding Protein 2 Inactivation on β -Lactamase Expression and Muropeptide Profile. PMID: 28861525 — PBP2 function and muropeptide effects.

Prepared for manual module satisfiability and gene-annotation curation. All KT2440 functional calls are homology-inferred (UniProt PE=3); species-transfer strength is stated per claim. Module verdict: satisfiable with revisions — 11 in-scope genes cover all five canonical steps, but the bucket must be de-contaminated (12 out-of-scope genes), EC-completed (pbpC, mrcB), dacB reclassified (PBP4/S13 D,D-endopeptidase), and extended with L,D-transpeptidation (PP_1451, PP_2320) and PBP4 endopeptidase nodes. The two MrdA/PBP2 paralogs are flagged candidate_uncertain and promoted to full gene review.
